Varun Pahuja

Jaipur, Rajasthan, India | +91 7999389919 | vpahuja1508@gmail.com
varunpahuja.vercel.app | linkedin.com/in/varunpahuja15 | github.com/VarunPahuja

Summary

Final-year CSE student who builds AI systems with deterministic controls, audit trails, and human approval. Backend lead on the AI-agent governance platform that won the Deloitte USI Capstone 2026; builder of an SMT-based payment-policy verifier and a beta RAG platform over government policy data. Comfortable turning ambiguous business requirements into documented, tested systems.

Technical Skills

- **Languages:** Python, SQL, TypeScript
- **Backend & Web:** FastAPI, Node.js, REST APIs, SQLAlchemy, Alembic, React, Next.js
- **Data & AI:** Pandas, NumPy, Scikit-learn, PyTorch, ETL Pipelines, LLM APIs (Gemini), LangGraph, RAG, Vector Search, Embeddings
- **Governance & Risk Controls:** Policy Engines, Z3/SMT Verification, prototype role-based authorization, Audit Logging, Deterministic Guardrails, Evaluation Harnesses
- **Databases & Cloud:** PostgreSQL/pgvector, MySQL, Supabase, Firebase, Docker, Vercel, Render
- **Engineering Practice:** Git/GitHub, CI/CD, Unit Testing (pytest), Technical Documentation (ADRs), Agile

Education

Manipal University Jaipur **Aug 2023 – May 2027**
B.Tech in Computer Science Engineering — 8.19 CGPA *Jaipur, Rajasthan*
– **Relevant Coursework:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Cloud Computing

Work Experience

To Let Globe **Jul 2025 – Oct 2025**
Founder's Office Intern *Remote*
– Translated business goals into technical briefs by evaluating competitor architectures, developer tooling, and AI capabilities; briefs informed product and engineering roadmap decisions
– Coordinated engineering and product deliverables across sprints, surfaced technical blockers to leadership, and tracked execution through Agile ceremonies
– Built and maintained technical and process documentation, consolidating engineering knowledge and improving onboarding workflows

Projects

Adaptive AI Governance Platform – Deloitte USI Capstone 2026 | FastAPI, PostgreSQL, LangGraph, Docker | Capstone Winner **2026**
- Owned the backend and policy engine for an invoice-agent governance platform with a five-rung limit ladder from Rs 500 to Rs 10,000, statistical evidence for increases, and automatic clawback when performance drifts.
- Enforced separation of duties: LLM agents recommend, statistics provide evidence, and the policy engine enforces; humans approve every autonomy increase, while decisions and policy changes are recorded in an audit log.
- Implemented prototype role-based authorization and PostgreSQL with Alembic migrations; coordinated a four-person team through frozen shared contracts and a CI-gated main branch.

Bounded – AI Agent Payment Policy Verifier | Python, Z3/SMT, FastAPI, Razorpay | Razorpay Buildathon **2026**
- Built a bounded-horizon SMT verifier for transaction caps, spending-window limits and refund invariants.
- Separated LLM policy interpretation from Z3 ALLOW/BLOCK checks; integrated fail-closed enforcement into Razorpay test mode.
- Pilot: 46 adversarial and benign scenarios with mocked Razorpay network calls; median Z3 verification latency 5.04 ms across 808 calls. Fixed-horizon prototype, not an unbounded or production safety guarantee.

Exit Plan – AI Immigration Platform | FastAPI, React, Supabase, Gemini API, pgvector | exit-plan-two.vercel.app May 2026
- Built and deployed a beta full-stack platform ranking 10 countries for Indian graduates using a weighted scoring engine.
- Implemented policy-data ingestion, document chunking and embeddings, with hybrid vector and keyword retrieval.
- Designed a retrieval workflow with source citations and a mock fallback when retrieval is unavailable; beta status does not establish production reliability or live policy-change alerts.

SustainBridge – AI Supply Chain Optimization | Python, Scikit-learn, Pandas | Demo **June 2025**
– Built an end-to-end demand forecasting pipeline; Random Forest with 5-fold CV reached **95.6% accuracy** and **0.92 F1**, while ETL optimizations cut runtime by **40%**
– Designed a stock redistribution algorithm reducing excess inventory by **30%**; built an admin dashboard for 500+ stores that cut lookup time by **40%**

Leadership & Activities

Glitch! eSports Society – Director of External Relations **May 2024 – April 2026**
Manipal University Jaipur
– Secured 20+ sponsors and raised Rs 200,000+ for LAN and esports events through stakeholder pitching and brand partnerships